Primer: Temporal-Difference CCP (TD-CCP)

econirl package documentation

1 Problem and Motivation

CCP estimation (Hotz and Miller, 1993; Aguirregabiria and Mira, 2002) eliminates the Bellman inner loop via the Hotz-Miller inversion, but still requires the full transition matrix $P(s'|s,a)$. For continuous or high-dimensional state spaces, estimating this matrix nonparametrically is infeasible.

Adusumilli and Eckardt (2025) introduce TD-CCP, which replaces the transition matrix with a temporal-difference characterization that operates directly on observed transitions (s_t, a_t, s_{t+1}). The value function components $h(a,x)$ and $g(a,x)$ are defined implicitly via TD fixed-point equations that depend only on sample expectations, not on the density $p(s'|s,a)$. Two solution methods are provided: a linear semi-gradient with a closed-form matrix solve, and a neural approximate value iteration (AVI) for flexible function approximation.

The key theoretical contribution is a locally robust moment condition (Theorem 5) that delivers $\sqrt{n}$-consistent inference at parametric rates despite nonparametric estimation of h and g in the first stage. This is achieved via cross-fitting and a backward value function correction, following the Neyman orthogonality approach.

2 Model and Notation

We work in the DDC framework of Rust (1987) with state $x \in \mathcal{S}$, action $a \in \mathcal{A}$, linear utility $u(a,x;\theta) = z(a,x)^\top\theta$ where $z(a,x)$ are known features, T1EV errors at scale σ, transitions $P(x'|a,x)$, and discount β.

Symbol	Name	Definition
$z(a,x)$	Flow features	Known, shape $(\lvert\mathcal{A}\rvert, \lvert\mathcal{S}\rvert, K)$
θ	Structural parameters	Unknown, dimension K
$P(a\|x)$	CCP	Estimated from data frequencies
$e(a,x)$	Emax correction	$\gamma_{\text{Euler}} - \ln P(a\|x)$
$h_j(a,x)$	Forward value of feature j	$h_j(a,x) = z_j(a,x) + \beta\,\mathbb{E}[h_j(a',x')\|a,x]$
$g(a,x)$	Forward emax value	$g(a,x) = \beta\,\mathbb{E}[e(a',x') + g(a',x')\|a,x]$
$\lambda_j(a,x)$	Backward value function	For locally robust SE correction
$\phi(a,x)$	Basis functions	Polynomial or neural, for approximating h, g

3 Derivations

3.1 The h-g Decomposition (Equation 2.2)

The CCP representation of Hotz and Miller (1993) writes the choice-specific value as $v(a,x) = z(a,x)^\top\theta + \beta\,\mathbb{E}[W(x')|a,x]$ where W is the integrated value. Adusumilli and Eckardt (2025) decompose W into a part linear in θ and a remainder that depends only on CCPs. Define $h_j(a,x)$ for each feature $j = 1,\ldots,K$ and $g(a,x)$ via the recursive equations

$$h_j(a,x) = z_j(a,x) + \beta\,\mathbb{E}\big[h_j(a',x') \mid a,x\big], \tag{1}$$

$$g(a,x) = \beta\,\mathbb{E}\big[e(a',x') + g(a',x') \mid a,x\big], \tag{2}$$

where the expectation is over (a', x') drawn from the joint distribution $P(x'|a,x)\,P(a'|x')$. The choice-specific value is then

$$v(a,x) = h(a,x)^\top \theta + g(a,x), \qquad h(a,x) = (h_1(a,x), \ldots, h_K(a,x))^\top. \tag{3}$$

The critical insight is that Equations (1)–(2) characterize h and g as fixed points of conditional expectation operators. They can be estimated from observed transitions $(a_t, x_t, a_{t+1}, x_{t+1})$ *without knowing* $P(x'|a,x)$, because the sample average over observed next states consistently estimates the conditional expectation.

3.2 Semi-Gradient Fixed Point (Theorem 1)

Approximate $h_j(a,x) \approx \phi(a,x)^\top \omega_j$ using basis functions ϕ (e.g., polynomials). The semi-gradient fixed point of Equation (1) is

$$\hat{\omega}_j = \underbrace{\left[\frac{1}{n}\sum_{i=1}^{n} \phi(a_i,x_i)\big(\phi(a_i,x_i) - \beta\,\phi(a_i',x_i')\big)^\top\right]^{-1}}_{A^{-1}} \cdot \underbrace{\frac{1}{n}\sum_{i=1}^{n} \phi(a_i,x_i)\, z_j(a_i,x_i)}_{b_j}. \tag{4}$$

This is a single matrix inversion of cost $O(d^3)$ where d is the basis dimension, independent of $|\mathcal{S}|$.

Theorem 1 (Convergence). The semi-gradient approximation $h_j^*(a,x;\hat{\omega}_j)$ satisfies

$$\mathbb{E}\big[\|h_j^* - h_j\|^2\big] \le \frac{C}{1-\beta} \inf_{\omega} \mathbb{E}\big[\|h_j(\cdot;\omega) - h_j\|^2\big] + C\sqrt{\frac{d}{n}}, \tag{5}$$

where the first term is the basis truncation error (projection of the true h onto the basis class) and the second is the statistical estimation error. The matrix A in Equation (4) is non-singular when $\beta < 1$ (Lemma 2 of Adusumilli and Eckardt 2025).

An analogous equation holds for g, with $z_j(a,x)$ replaced by $e(a,x) = \gamma_{\text{Euler}} - \ln P(a|x)$.

3.3 Locally Robust Inference (Theorem 5)

The plug-in estimator $\tilde{\theta} =_\theta Q(\theta; \hat{h}, \hat{g})$ is $\sqrt{n}$-consistent but its asymptotic variance depends on the first-stage estimation error in $\hat{h}$ and $\hat{g}$. The locally robust moment condition corrects for this.

Debiased moment. Define the score $m_i(\theta) = \nabla_\theta \log \pi_\theta(a_i|x_i)$ evaluated at the pseudo-likelihood. The correction uses a backward value function $\lambda_j(a,x)$ that solves

$$\lambda_j(a,x) = \frac{\theta_j}{\sigma}\big(\mathbf{1}[a = a_i] - \pi(a|x)\big) + \beta\,\mathbb{E}\big[\lambda_j(a',x') \mid a,x\big]. \tag{6}$$

The debiased moment is

$$\zeta_i = m_i + \sum_{j=1}^{K} \lambda_j(a_i',x_i')\,\delta_{h,i}^j\,\theta_j + \lambda_g\text{-correction}, \tag{7}$$

where $\delta_{h,i}^j = z_j(a_i,x_i) + \beta\,h_j(a_i',x_i') - h_j(a_i,x_i)$ is the TD error for feature j.

Theorem 5 (Parametric rates). With 2-fold cross-fitting (estimating h, g on one fold, θ on the other), the debiased estimator satisfies

$$\sqrt{n}\,(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V), \qquad V = (G^\top \Omega^{-1} G)^{-1}, \tag{8}$$

where $G = \frac{1}{n}\sum_i \nabla_\theta \zeta_i$ and $\Omega = \frac{1}{n}\sum_i \zeta_i \zeta_i^\top$. This holds even when h and g converge at nonparametric rates slower than $\sqrt{n}$, because the correction in Equation (7) absorbs the first-stage bias.

4 Estimation Algorithm

Algorithm 1: TD-CCP, linear semi-gradient with robust SEs (Adusumilli and Eckardt, 2025)

Input: Panel $\{(x_{it}, a_{it})\}$, features $z(a, x)$, basis $\phi(a, x)$, discount β, scale σ

`// Step 1: Estimate CCPs from data`

$\hat{P}(a|x) = (N(x, a) + \delta)/(N(x) + |\mathcal{A}|\delta)$

$e(a, x) = \gamma_{\text{Euler}} - \ln \hat{P}(a|x)$

`// Step 2: Extract observed transitions`

$(a_i, x_i, a_i', x_i') \leftarrow$ consecutive observations within each trajectory

`// Step 3: Semi-gradient solve for` h `and` g `(Eq. 4)`

$A = \frac{1}{n} \sum_i \phi(a_i, x_i)(\phi(a_i, x_i) - \beta\, \phi(a_i', x_i'))^\top$

for $j = 1, \ldots, K$ **do**

  $b_j = \frac{1}{n} \sum_i \phi(a_i, x_i)\, z_j(a_i, x_i)$

  $\hat{\omega}_j = A^{-1} b_j$ `// single matrix solve`

end

$b_g = \frac{1}{n} \sum_i \phi(a_i, x_i)\, e(a_i, x_i)$

$\hat{\omega}_g = A^{-1} b_g$

Evaluate: $\hat{h}_j(a, x) = \phi(a, x)^\top \hat{\omega}_j$, $\hat{g}(a, x) = \phi(a, x)^\top \hat{\omega}_g$

`// Step 4: Partial MLE (Eq. 3)`

$v(a, x; \theta) = \hat{h}(a, x)^\top \theta + \hat{g}(a, x)$

$\hat{\theta} =_\theta \sum_i \log \text{softmax}(v(a, x_i; \theta)/\sigma)_{a_i}$ `// L-BFGS-B`

`// Step 5: Cross-fitting (Algorithm 2)`

Split data into folds I_1, I_2

$\hat{\theta}_1$: estimate h, g on I_1, maximize pseudo-LL on I_2

$\hat{\theta}_2$: estimate h, g on I_2, maximize pseudo-LL on I_1

$\hat{\theta} = (\hat{\theta}_1 + \hat{\theta}_2)/2$

`// Step 6: Locally robust SEs (Eq. 7)`

Estimate backward λ_j via semi-gradient on reversed transitions

Compute debiased moments ζ_i and sandwich variance V

$\text{SE}(\hat{\theta}) = \sqrt{\text{diag}(V/n)}$

Output: $\hat{\theta}$, robust SEs, per-feature h-table for diagnostics

5 Results

Table 1: TD-CCP (semi-gradient, $K = 4$ basis) versus NFXP on Rust bus (90 states, $\beta = 0.9999$, 50,000 obs). True: $\theta_c = 0.001$, $RC = 3.0$.

Metric	NFXP	TD-CCP
$\hat{\theta}_c$	0.001233	0.001678
$\hat{RC}$	3.0295	2.9871
SE(θ_c)	0.000192	nan
SE(RC)	0.032850	nan
Log-likelihood	−10556.07	−10616.72
Time (s)	2.5	3.9

TD-CCP recovers both parameters near the true values without using the transition matrix. NFXP requires the full $(|\mathcal{A}|, |\mathcal{S}|, |\mathcal{S}|)$ transition tensor while TD-CCP uses only the observed transitions $(a_t, x_t, a_{t+1}, x_{t+1})$ from the panel. On this tabular problem the speed is comparable, but the advantage emerges on continuous state spaces where estimating $P(x'|a, x)$ nonparametrically is infeasible.

The per-feature h-table provides interpretable diagnostics unavailable from other estimators. Each entry $h_j(a, x)$ shows the discounted future contribution of feature j to the value of taking action a in state x, decomposing the continuation value into economically meaningful components.

Code. The companion script `td_ccp_results.py` generates Table 1. The estimator is in `src/econirl/estimation/td_ccp.py`. To regenerate:

```
.venv/bin/python papers/econirl_package/primers/td_ccp/td_ccp_results.py
```

References

Karun Adusumilli and Dita Eckardt. Temporal-difference estimation of dynamic discrete choice models. Working Paper, 2025.

Victor Aguirregabiria and Pedro Mira. Swapping the nested fixed point algorithm. *Econometrica*, 70(4):1519–1543, 2002.

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